The Cox–Ross–Rubinstein Binomial Model: Convergence, Oscillation, and Calibration

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Abstract

This note presents a self-contained derivation of the Cox–Ross–Rubinstein (1979) binomial option pricing model and its convergence to the Black–Scholes formula. Great emphasis is placed on giving detailed proofs of the no-arbitrage pricing relations and deriving the risk-neutral measure through two different approaches. Additionally, we prove the convergence of the discrete-time model to its continuous-time limit using the Lindeberg–Feller central limit theorem. We then proceed to studying the non-monotone oscillation phenomenon characteristic of the standard CRR method and discuss the different smoothing alternatives of Tian (1999) and Leisen and Reimer (1996). The theoretical results are accompanied by a thorough numerical implementation in Python and C++ and an empirical calibration study using S&P 500 index option data from OptionMetrics and Wharton Research Data Services (WRDS). Several interesting features of the convergence behavior are discussed.

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1 Introduction

Option pricing theory changed fundamentally with the works of Black, Scholes, and Merton in the early 1970s. Their continuous-time framework produced the well-known and celebrated Black-Scholes formula

$$C^{\mathrm{BS}}(S_0, K, T, r, \sigma) = S_0 N(d_1) - K e^{-rT} N(d_2),$$

which gives the arbitrage-free value of a European call option in terms of the spot price S_0, strike K, time to expiration T, risk-free rate r, and the volatility σ.

The original derivation, however, relies on continuous-time stochastic calculus, Itô's formula, and partial differential equations—tools not available to economists and finance practitioners at the time. While mathematically elegant, this advanced framework can obscure the underlying economic structure of the pricing argument.

A few years later, Cox, Ross, and Rubinstein (1979) introduced a discrete-time alternative based on a multiplicative binomial tree. In their model, the stock price moves up or down by fixed factors at each trading period (random walk), and option values are obtained through replication and backward induction. As the number of periods N tends to infinity, the binomial price converges to the Black–Scholes price.

Today, the CRR binomial model remains one of the standard introductions to arbitrage-free pricing because it isolates the main ideas of the theory in an elementary one-period setting: replication, no-arbitrage pricing, risk-neutral probabilities, and martingale methods all appear in a finite-state form before moving to continuous-time pricing.

This note develops the CRR model from first principles, followed by a study of its convergence to Black–Scholes, both mathematically and empirically. We give particular attention to the oscillatory convergence behavior of the standard CRR tree and to the smoothing schemes introduced by Tian (1999) and Leisen–Reimer (1996) and its effects on oscillation behavior.

1.1 Scope and Contribution

This mathematical note presents a self-contained treatment of the Cox-Ross-Rubinstein binomial model and its convergence to the Black-Scholes formula.

The exposition focuses on three main points. First, we derive the discrete-time pricing framework directly from replication and no-arbitrage conditions, showing how the risk-neutral measure and martingale pricing emerge in a finite-state setting. Second, we give a mathematically detailed proof of convergence to Black–Scholes using the Lindeberg–Feller central limit theorem for triangular arrays (Theorem 5.5). Third, we study the oscillatory convergence behavior of the standard CRR scheme and discuss the smoothing approaches of Tian (1999) and Leisen–Reimer (1996).

I aim for the level of rigor appropriate to a graduate-level expository note: every theorem is stated precisely, every proof is given in detail or referenced explicitly, and every numerical claim is grounded either in demonstrated theory or in computations available in the accompanying coding implementation. The exposition draws particularly on the original CRR paper, on Shreve's (2004) treatment in *Stochastic Calculus for Finance II*, and on the recent classroom

notes of Chen (2016, 2018) and Chen and Chen (2025) on elementary derivations of the constants appearing in the Black–Scholes formula.

The note's primary content lies in three contributions. The first is a detailed proof of CRR's convergence theorem (Theorem 5.5) via the Lindeberg–Feller central limit theorem for triangular arrays, with the explicit verification of every step that is shown in the original paper. The second is a detailed exposition of the oscillation phenomenon (Section 6): the well-documented but often under-discussed observation that CRR convergence to Black–Scholes is not monotone in N, exhibiting period-2 oscillations with an $O(1/N)$ envelope. We also discuss the mechanism (the misalignment between the strike K and the terminal binomial nodes), state carefully the asymptotic expansion of Diener and Diener (2004), and review the smoothing alternatives of Tian (1999) and Leisen and Reimer (1996).

Note 1: I make no claim of original mathematical results or findings. In fact, the CRR model and its convergence are foundational and well-known; the oscillation phenomenon and its asymptotic expansion are widely used in the literature; and the Tian and Leisen–Reimer schemes are standard tools used by practitioners in the space. The contribution of this note is thus expository: a unified, self-contained, and mathematically careful presentation of main derivatives topics in a single document, with proofs filled in where standard references skip them, and with numerical and empirical evidence provided alongside the theory. This last part is where the main derivations come to life.

Note 2: The treatment of Tian (1999) in Section 6.4 is a revision of an earlier draft. The accompanying coding implementation now follows the construction of Tian (1999) eqs. (6) and (13) directly, verified against the paper's Tables I and II to four-to-six decimal places, with the smoothing Richardson extrapolation of eq. (17) provided as a separate resource. Later, we will observe the implementation of both methods and their different behavior when it comes to pricing error estimation.

Note 3 (methodology): The mathematical derivations in this note are the author's own; complete handwritten worksheets for the principal derivations (one-period replication, the closed-form CRR formula, CRR parameter asymptotics, the mean and variance of Z_N, the Lindeberg–Feller verification, binomial tail convergence, stock-numeraire convergence, and the oscillation mechanism) are available in the project repository. AI-assisted tools were used for prose polish, LaTeX typesetting, and figure-script structure.

The handwritten derivations underlying Sections 2.2, 3.3, 5.1, 5.4, 5.5, 5.6, 5.7, and 6.2 are available in the project repository at notes/CRR_Handwritten_Derivations.pdf.

1.2 Connection to Earlier Work in This Series

The constants $\sqrt{2\pi}$ and the standard normal CDF (Cumulative Distribution Function) that appear in the Black–Scholes formula have, in fact, several routes to elementary derivation. Newton's semicircle produces a geometric construction of the Gaussian integral, presented in Chen (2016); the bounds $(1 - x/n)^n \leq e^{-x} \leq (1 + x/n)^{-n}$, applied to the probability integral, yield Wallis's product and the $\sqrt{\pi}/2$ value, presented in Chen (2018); and Taylor's formula with integral remainder produces the beta-gamma function identity, which presented in Chen and Chen (2025). Each derivation uses only single-variable calculus, thus avoiding the multivariable

polar-coordinate argument typical of textbook methods. The convergence proof of this note connects to that program in Section 5.9: the discrete-to-continuous limit established by the Lindeberg–Feller theorem is, by its nature, the statement that binomial tail probabilities converge to standard normal tail probabilities, and the constant $\sqrt{2\pi}$ that normalizes the Gaussian density is the same constant that those elementary derivations evaluate. The present note may therefore be read as a discrete-time alternative to the analytic derivations in that mentioned sequence. The same constants appear, but they emerge from a probabilistic limit theorem rather than from analytic identities.

1.3 Organization

Section 2 defines and develops the one-period two-state model, deriving the unique no-arbitrage price of a contingent claim and the risk-neutral probability. Section 3 extends the same analysis to N periods, deriving the closed-form CRR pricing formula for European calls through backward induction. Section 4 deals with American options and the optimal-stopping problem, including the early-exercise boundary and the adjustments required to handle dividends. Section 5 contains the main mathematical result: a complete proof that the CRR price converges to the Black–Scholes formula as $N \to \infty$, via the Lindeberg–Feller central limit theorem. Section 6 characterizes the oscillation phenomenon and reviews the smoothing schemes of Tian and Leisen–Reimer. Section 7 presents algorithmic details and convergence experiments. Section 8 concludes and discusses planned extensions of this work.

1.4 Notation and Conventions

It is important to note that we adopt the following conventions throughout. Time is denoted t in continuous-time contexts and n in discrete-time contexts; horizons are T and N respectively, with $\Delta t = T/N$. The risk-free gross return is $R = e^{r\Delta t}$ over one period in the discrete-time setting and e^{rT} over the full horizon T in the continuous-time setting. The physical (real-world) probability measure is $\mathbb{P}$; the risk-neutral measure is $\mathbb{Q}$, and expectations under each are denoted $\mathbb{E}$ and $\mathbb{E}^{\mathbb{Q}}$ respectively. The standard normal CDF is $N(\cdot)$, in keeping with finance convention; we use N for the number of binomial periods, but it is clarified to avoid confusion depending on the context. The complementary binomial distribution function is denoted $\Phi(a; N, p)$, defined in equation (15), and must not be confused with the standard normal CDF $N(\cdot)$. Throughout, log means natural log.

2 The One-Period Binomial Model

Let us begin the simplest non-trivial setting. Suppose we have a market with a single risky asset, a risk-free bond (yielding the risk-free rate), and a single period of trading. Despite its simplicity, this model contains every essential idea of arbitrage-free option pricing. The risk-neutral measure, the replicating portfolio, and the independence of option value from physical probabilities all begin here. The multi-period and continuous-time models that follow this setting are, in a precise sense, iterated and limiting versions of the same idea.

2.1 The Market

Fix a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ with two states, $\Omega = \{\omega_u, \omega_d\}$, equipped with the physical (or *real-world*) probability measure $\mathbb{P}$ assigning $\mathbb{P}(\omega_u) = q$ and $\mathbb{P}(\omega_d) = 1 - q$ for some $q \in (0, 1)$. The market consists of two assets traded at times $t = 0$ and $t = 1$:

1. A *risk-free bond* with price $B_0 = 1$ at time 0 and $B_1 = R$ at time 1, where $R = 1 + r_f > 0$ and r_f denotes the one-period risk-free rate. The bond's terminal value is deterministic.

2. A *risky asset* ("stock") with price $S_0 = S > 0$ at time 0. Its time-1 price S_1 takes two values:
$$S_1(\omega) = \begin{cases} uS & \text{if } \omega = \omega_u, \\ dS & \text{if } \omega = \omega_d, \end{cases}$$
where $0 < d < u$ are the gross returns in the two states.

We must impose throughout the no-arbitrage condition

$$d < R < u. \tag{1}$$

Its necessity is established in the following lemma.

Lemma 2.1 (No-arbitrage condition)**.** *The one-period market admits no arbitrage if and only if $d < R < u$.*

Proof. Suppose first that $R \leq d$. Then borrowing one dollar at the risk-free rate and investing in the stock yields a portfolio with time-1 value $S_1 - R \geq dS - R \geq 0$ in every state, with strict inequality in state ω_u since $u > d \geq R$. This is an arbitrage. Symmetrically, if $R \geq u$, shorting the stock and investing the proceeds in the bond produces an arbitrage. Conversely, suppose $d < R < u$. Any portfolio (Δ, B) consisting of Δ shares and B dollars in the bond has time-0 value $\Delta S + B$ and time-1 values $\Delta uS + RB$ and $\Delta dS + RB$. If both terminal values are non-negative and at least one is positive, then $\Delta S + B > 0$, as may be verified by considering the cases $\Delta \geq 0$ and $\Delta < 0$ separately and using $d < R < u$. Hence no arbitrage exists. $\square$

2.2 Contingent Claims and Replication

A *contingent claim* maturing at time 1 is any random variable $X : \Omega \to \mathbb{R}$, specified by its payoffs $X_u := X(\omega_u)$ and $X_d := X(\omega_d)$. The European call option with strike K corresponds to $X = (S_1 - K)^+$, so $X_u = (uS - K)^+$ and $X_d = (dS - K)^+$. The European put corresponds to $X = (K - S_1)^+$.

Definition 2.2 (Replicating portfolio)**.** A *replicating portfolio* for a claim X is a pair $(\Delta, B) \in \mathbb{R}^2$ satisfying

$$\Delta \cdot uS + RB = X_u, \qquad \Delta \cdot dS + RB = X_d. \tag{2}$$

The system (2) is linear in (Δ, B) with determinant $(u - d)RS \neq 0$, so a unique replicating portfolio exists for every claim. Solving:

$$\Delta = \frac{X_u - X_d}{(u - d)S}, \qquad B = \frac{1}{R} \cdot \frac{uX_d - dX_u}{u - d}. \tag{3}$$

The quantity Δ is the *hedge ratio*: the number of shares required to replicate one unit of the claim.

Theorem 2.3 (One-period pricing formula)**.** *Under the no-arbitrage condition* (1)*, the unique arbitrage-free time-*0 *price of any contingent claim* X *is*

$$V_0 = \Delta S + B = \frac{1}{R}\left[p\, X_u + (1-p)\, X_d\right], \tag{4}$$

where

$$p := \frac{R-d}{u-d}. \tag{5}$$

Proof. Suppose the claim trades at price $V_0' \neq \Delta S + B$. If $V_0' > \Delta S + B$, sell the claim, buy the replicating portfolio, and invest the difference $V_0' - (\Delta S + B) > 0$ in the bond; at time 1 the portfolio's payoff exactly cancels the claim's, leaving a riskless profit of $R(V_0' - (\Delta S + B)) > 0$. If $V_0' < \Delta S + B$, reverse the position. Either case contradicts no-arbitrage, so $V_0 = \Delta S + B$. Substituting (3) and simplifying:

$$\Delta S + B = \frac{X_u - X_d}{u-d} + \frac{1}{R} \cdot \frac{uX_d - dX_u}{u-d} = \frac{1}{R}\left[\frac{R(X_u - X_d) + uX_d - dX_u}{u-d}\right].$$

The numerator equals $(R-d)X_u + (u-R)X_d$, giving (4) with p as in (5). □

By condition (1), $p \in (0,1)$. We may therefore interpret p as a probability and define a new measure $\mathbb{Q}$ on Ω by $\mathbb{Q}(\omega_u) = p$, $\mathbb{Q}(\omega_d) = 1-p$. Under $\mathbb{Q}$, the expected gross return on the stock equals R:

$$\mathbb{E}^{\mathbb{Q}}[S_1] = p \cdot uS + (1-p) \cdot dS = S \cdot \frac{(R-d)u + (u-R)d}{u-d} = RS.$$

We call $\mathbb{Q}$ the *risk-neutral measure* (or *equivalent martingale measure*, since the discounted price S_t/R^t is a $\mathbb{Q}$-martingale). The pricing formula (4) thus reads

$$V_0 = \frac{1}{R}\mathbb{E}^{\mathbb{Q}}[X], \tag{6}$$

the discounted risk-neutral expectation of the payoff. Three features of this result deserve emphasis.

Remark 2.4. The physical probability q does not appear in (4). Investors who disagree about the likelihood of upward and downward moves nonetheless agree on the option's price. Pricing is determined entirely by the absence of arbitrage and the no-arbitrage parameters u, d, R.

Remark 2.5. The derivation makes no assumption about investor preferences beyond non-satiation (preferring more wealth to less, in order to exploit arbitrage). In particular, no utility function or risk-aversion parameter enters the pricing formula.

Remark 2.6. Although $\mathbb{Q}$ is constructed from the no-arbitrage parameters rather than from any economic primitive, it admits a natural interpretation: $\mathbb{Q}$ is the unique probability measure equivalent to $\mathbb{P}$ under which the discounted stock price is a martingale. This martingale property generalizes to the multi-period and continuous-time settings and underlies the modern formulation of arbitrage pricing due to Harrison and Kreps (1979) and Harrison and Pliska (1981).

2.3 An Alternative Derivation via Binary Claims

The replicating-portfolio argument above is standard but, as Orosi (2015) observes, it bundles two distinct ideas: the existence of a replicating portfolio and the resulting risk-neutral probability. The following alternative derivation isolates the second idea by pricing a single elementary claim, the *binary call*, and recovering p directly. We include it both for its pedagogical clarity and because the binary-claim viewpoint generalizes cleanly to the multi-period setting.

Let $\mathbf{1}_u$ denote the binary claim paying one unit in state ω_u and zero in state ω_d. By Theorem 2.3 with $X_u = 1$, $X_d = 0$, its price is

$$V_0(\mathbf{1}_u) = \frac{1}{R} \cdot \frac{R-d}{u-d} = \frac{p}{R}. \tag{7}$$

The price p/R has a transparent interpretation: it is the discounted probability of the up state under $\mathbb{Q}$. By symmetry, $V_0(\mathbf{1}_d) = (1-p)/R$. Any claim X may be decomposed as $X = X_u\mathbf{1}_u + X_d\mathbf{1}_d$, and linearity of pricing yields (4) immediately. The binary claims $\mathbf{1}_u, \mathbf{1}_d$ are sometimes called *Arrow–Debreu securities* for the two states; their prices, p/R and $(1-p)/R$, are the state prices.

2.4 An Example

Following Cox, Ross, and Rubinstein (1979), consider a stock with $S = 50$, $u = 2$, $d = 1/2$, and a one-period risk-free rate giving $R = 5/4$. A European call with strike $K = 50$ has payoffs $C_u = 50$ and $C_d = 0$. Then

$$p = \frac{R-d}{u-d} = \frac{5/4 - 1/2}{2 - 1/2} = \frac{1}{2},$$

and the call's price is

$$C_0 = \frac{1}{R}\big[p \cdot 50 + (1-p) \cdot 0\big] = \frac{4}{5} \cdot 25 = 20.$$

The replicating portfolio holds $\Delta = (50-0)/((2-1/2)\cdot 50) = 2/3$ shares and borrows $B = -(1/R)(dC_u - uC_d)/(u-d) = -(4/5)(25/(3/2)) = -40/3$. One verifies directly that this portfolio delivers 50 in state ω_u and 0 in state ω_d, matching the call's payoff in each state.

3 The N-Period Binomial Model

We now extend the one-period model to a finite-horizon discrete-time setting with N trading periods. The underlying logic is unchanged: at each node of the resulting tree, the one-period analysis of Section 2 applies locally. What is new is the *recursive* structure of pricing, the closed-form expression for European claims, and the explicit appearance of the binomial distribution in the pricing formula.

3.1 The Multiplicative Binomial Tree

Fix a horizon of N periods, each of length $\Delta t > 0$. The stock price evolves on a recombining binomial tree: at each node, the price either multiplies by u (with $\mathbb{P}$-probability q) or by d (with $\mathbb{P}$-probability $1-q$), independently across periods. After j upward moves and $N-j$ downward

moves, the time-N price equals $u^j d^{N-j} S_0$. Because the tree recombines—i.e., $ud = du$—the order of the moves is immaterial, and the time-N price depends only on the total number of upward moves $J_N \in \{0, 1, \ldots, N\}$.

We retain the one-period risk-free gross return $R = 1 + r_f > 0$, constant across periods, and the no-arbitrage condition $d < R < u$. Let $\mathbb{Q}$ denote the measure under which $\mathbb{Q}(\text{up}) = p$ and $\mathbb{Q}(\text{down}) = 1 - p$ at each node, with

$$p = \frac{R - d}{u - d} \in (0, 1)$$

as in (5). Under $\mathbb{Q}$, the upward-move counts J_N follow a binomial distribution with parameters N and p.

3.2 Recursive Pricing

Let $V_n(j)$ denote the time-n price of a European claim X at the node reached by j upward moves out of n, for $n = 0, 1, \ldots, N$ and $j = 0, 1, \ldots, n$. At maturity,

$$V_N(j) = X(u^j d^{N-j} S_0), \qquad j = 0, 1, \ldots, N. \tag{8}$$

At any earlier node, the one-period pricing formula (4) applied locally yields

$$V_n(j) = \frac{1}{R}\big[p\, V_{n+1}(j+1) + (1-p)\, V_{n+1}(j)\big]. \tag{9}$$

Equation (9) is the *backward induction* or *dynamic programming* relation for European claims. Together with (8), it determines $V_n(j)$ at every node, and in particular determines the time-0 price $V_0 = V_0(0)$.

Theorem 3.1 (N-period European pricing formula)**.** *The arbitrage-free time-0 price of a European claim X with maturity N is*

$$V_0 = \frac{1}{R^N} \sum_{j=0}^{N} \binom{N}{j} p^j (1-p)^{N-j}\, X(u^j d^{N-j} S_0). \tag{10}$$

Proof. We prove (10) by induction on N. The case $N = 1$ is Theorem 2.3. Suppose (10) holds for some horizon $N \geq 1$, and consider a claim with horizon $N + 1$. Applying (9) at time zero and the inductive hypothesis to the two subtrees rooted at $(1, 1)$ and $(1, 0)$:

$$V_0 = \frac{1}{R}\big[p\, V_1(1) + (1-p)\, V_1(0)\big],$$

where, by hypothesis,

$$V_1(1) = \frac{1}{R^N} \sum_{k=0}^{N} \binom{N}{k} p^k (1-p)^{N-k}\, X(u^{k+1} d^{N-k} S_0),$$
$$V_1(0) = \frac{1}{R^N} \sum_{k=0}^{N} \binom{N}{k} p^k (1-p)^{N-k}\, X(u^k d^{N-k+1} S_0).$$

Substituting and reindexing the first sum with $j = k + 1$:

$$V_0 = \frac{1}{R^{N+1}} \sum_{j=0}^{N+1} c_j \, X(u^j d^{N+1-j} S_0),$$

where, for $1 \le j \le N$,

$$c_j = p \cdot \binom{N}{j-1} p^{j-1}(1-p)^{N-j+1} + (1-p) \cdot \binom{N}{j} p^j (1-p)^{N-j} = \binom{N+1}{j} p^j (1-p)^{N+1-j},$$

using Pascal's identity $\binom{N+1}{j} = \binom{N}{j-1} + \binom{N}{j}$. The boundary cases $j = 0$ and $j = N + 1$ are verified directly. This yields (10) for horizon $N + 1$. □

Equivalently, in the notation of risk-neutral expectation,

$$V_0 = \frac{1}{R^N} \mathbb{E}^{\mathbb{Q}}[X(S_N)]. \tag{11}$$

3.3 The Closed-Form Call Pricing Formula

For a European call with strike K, the payoff is $X(s) = (s - K)^+$. Substituting into (10):

$$C_0 = \frac{1}{R^N} \sum_{j=0}^{N} \binom{N}{j} p^j (1-p)^{N-j} \, (u^j d^{N-j} S_0 - K)^+. \tag{12}$$

The summand is positive precisely when $u^j d^{N-j} S_0 > K$, equivalently

$$j \log(u/d) > \log(K/(d^N S_0)),$$

or $j > \log(K/(d^N S_0))/\log(u/d)$, since $\log(u/d) > 0$. Define

$$a := \min \left\{ j \in \{0, 1, \ldots, N+1\} : u^j d^{N-j} S_0 > K \right\}, \tag{13}$$

the smallest non-negative integer j for which the option finishes in the money. (If no such $j \le N$ exists, the option expires worthless and $C_0 = 0$; we set $a = N + 1$ in this case.) Then (12) becomes

$$C_0 = \frac{1}{R^N} \sum_{j=a}^{N} \binom{N}{j} p^j (1-p)^{N-j} (u^j d^{N-j} S_0 - K). \tag{14}$$

We split the sum into stock and strike components. Define the complementary binomial distribution function

$$\Phi(a; N, \pi) := \sum_{j=a}^{N} \binom{N}{j} \pi^j (1-\pi)^{N-j}, \tag{15}$$

the probability that a Binomial(N, π) random variable takes a value of at least a. The strike-component sum in (14) is precisely $\Phi(a; N, p) \cdot K/R^N$. For the stock component, define

$$p' := \frac{u}{R} \cdot p. \tag{16}$$

We claim $p' \in (0,1)$. Positivity is immediate. For $p' < 1$, observe that $1 - p' = (d/R)(1-p)$, since

$$\frac{u}{R}p + \frac{d}{R}(1-p) = \frac{up + d(1-p)}{R} = \frac{u(R-d) + d(u-R)}{R(u-d)} = \frac{R(u-d)}{R(u-d)} = 1.$$

Thus $1 - p' = (d/R)(1-p) > 0$, so $p' < 1$. Now the stock-component sum satisfies

$$\begin{aligned}\frac{1}{R^N}\sum_{j=a}^{N}\binom{N}{j}p^j(1-p)^{N-j}\,u^j d^{N-j}S_0 &= S_0\sum_{j=a}^{N}\binom{N}{j}\left(\frac{up}{R}\right)^j\left(\frac{d(1-p)}{R}\right)^{N-j}\\ &= S_0\sum_{j=a}^{N}\binom{N}{j}(p')^j(1-p')^{N-j} = S_0\,\Phi(a;N,p').\end{aligned}$$

Combining, we obtain the binomial call pricing formula.

Theorem 3.2 (Cox–Ross–Rubinstein, 1979)**.** *The arbitrage-free time-*0 *price of a European call option with strike K and maturity N in the binomial model is*

$$C_0 = S_0 \cdot \Phi(a;N,p') - \frac{K}{R^N}\cdot\Phi(a;N,p), \tag{17}$$

where p, p', Φ, and a are defined in (5)*,* (16)*,* (15)*, and* (13) *respectively.*

Remark 3.3. The formula (17) mirrors the structure of the Black–Scholes formula, $C_0 = S_0N(d_1) - Ke^{-rT}N(d_2)$, with the binomial tail probabilities $\Phi(a;N,p')$ and $\Phi(a;N,p)$ playing the role of $N(d_1)$ and $N(d_2)$. This is no accident: as $N \to \infty$ under the CRR parameterization $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$, the binomial tails converge to the corresponding normal tails. The proof of this convergence is the subject of Section 5.

Remark 3.4. The probability p' admits a measure-change interpretation. Under the measure $\mathbb{Q}'$ defined by $\mathbb{Q}'(\text{up}) = p'$, the discounted process S_t/R^t is no longer a martingale; instead, $K \cdot R^{-t}$ becomes the natural numeraire-discounted strike. Equivalently, p' is the risk-neutral probability of an upward move under the change of numeraire from the bond to the stock. This stock-numeraire (or *share measure*) interpretation generalizes to continuous time and provides one route to deriving the Black–Scholes formula via the Girsanov theorem.

3.4 Put Options and Put–Call Parity

For a European put with strike K, the payoff is $(K-s)^+$. The same analysis yields

$$P_0 = \frac{K}{R^N}\cdot\big[1-\Phi(a;N,p)\big] - S_0\cdot\big[1-\Phi(a;N,p')\big]. \tag{18}$$

Equivalently, by direct manipulation or by the model-free *put–call parity* relation

$$C_0 - P_0 = S_0 - \frac{K}{R^N}, \tag{19}$$

which holds for European options on non-dividend-paying assets in any arbitrage-free model. Parity follows from observing that a portfolio long one call and short one put has terminal payoff $S_N - K$ in every state, which is replicated by a long stock position and a short bond position of face value K.

4 American Options and Early Exercise

The pricing theory developed thus far applies to *European* claims, which can be exercised only at maturity. However, many traded options are *American*, granting the holder the right to exercise at any time up to and including maturity. The additional flexibility of American exercise introduces a fundamentally new mathematical structure: pricing becomes an *optimal stopping problem*, and closed-form expressions of the type (17) are generally unavailable. The binomial model, however, accommodates American exercise with only a minor modification of the backward induction (9), which accounts for much of its enduring practical importance.

4.1 The Early Exercise Premium

Consider an American put with strike K on a non-dividend-paying stock, with one period to maturity. Its time-0 holder may either exercise immediately, receiving $(K - S_0)^+$, or hold the option and receive its European value

$$P_0^{\mathrm{EU}} = \frac{1}{R}\big[p\,(K - uS_0)^+ + (1-p)\,(K - dS_0)^+\big].$$

The American value equals the larger:

$$P_0^{\mathrm{AM}} = \max\{(K - S_0)^+,\, P_0^{\mathrm{EU}}\}. \tag{20}$$

Whether immediate exercise is optimal depends on the parameters. If $dS_0 < K < uS_0$, then $(K - uS_0)^+ = 0$ and $P_0^{\mathrm{EU}} = (1-p)(K - dS_0)/R$. Immediate exercise dominates when $K - S_0 > (1-p)(K - dS_0)/R$, equivalently

$$R(K - S_0) > (1-p)(K - dS_0). \tag{21}$$

For sufficiently small S_0 (deep in the money), the left side grows linearly in $K - S_0$ while the right side approaches $(1-p)K$ in the limit, so (21) eventually holds. There exists, therefore, a critical stock price $\hat{S}$ below which immediate exercise is optimal.

For an American call on a non-dividend-paying stock, the situation is different.

Proposition 4.1 (Merton, 1973)**.** *For a European call on a non-dividend-paying stock with strike K, maturity N, and gross risk-free return $R > 1$ per period,*

$$C_n^{\mathrm{EU}} \geq S_n - \frac{K}{R^{N-n}} > S_n - K \quad \textit{at every node } (n, j).$$

Consequently, $C_n^{\mathrm{EU}} \geq (S_n - K)^+$ strictly, and an American call on a non-dividend-paying stock should never be exercised before maturity. Its price equals that of the corresponding European call.

Proof. By risk-neutral pricing,

$$C_n^{\mathrm{EU}} = \frac{1}{R^{N-n}}\mathbb{E}^{\mathbb{Q}}\big[(S_N - K)^+ \mid \mathcal{F}_n\big] \geq \frac{1}{R^{N-n}}\mathbb{E}^{\mathbb{Q}}[S_N - K \mid \mathcal{F}_n] = S_n - \frac{K}{R^{N-n}},$$

where the last equality uses the $\mathbb{Q}$-martingale property of the discounted stock price. Since $R > 1$, $K/R^{N-n} < K$ for $n < N$, so $C_n^{\mathrm{EU}} > S_n - K$, and a fortiori $C_n^{\mathrm{EU}} > (S_n - K)^+$. Exercising at time $n < N$ yields $(S_n - K)^+$, strictly less than the option's continuation value; exercise is therefore suboptimal. □

The proposition reveals an asymmetry: puts may be exercised early because the holder forfeits future interest on the strike K by waiting, whereas calls on non-dividend-paying stocks should be held because the holder *benefits* from delaying the strike payment. When the underlying pays dividends, this asymmetry is partially restored: large dividends can make early call exercise optimal, since holding the call (rather than the stock) forfeits the dividend.

4.2 The Optimal Stopping Formulation

Let $\mathcal{T}_n$ denote the set of stopping times τ taking values in $\{n, n+1, \ldots, N\}$, adapted to the filtration generated by the binomial tree. The American option pricing problem is

$$V_n^{\mathrm{AM}} = \sup_{\tau \in \mathcal{T}_n} \mathbb{E}^{\mathbb{Q}}\left[\frac{X(S_\tau)}{R^{\tau-n}} \,\middle|\, \mathcal{F}_n\right], \tag{22}$$

where $X(\cdot)$ is the payoff function of the option (e.g., $X(s) = (K-s)^+$ for a put). The supremum is attained because $\mathcal{T}_n$ is finite. The classical theory of optimal stopping (Snell 1952) yields the following *Snell envelope* characterization, which we state in the binomial setting.

Theorem 4.2 (Snell envelope)**.** *The American value process V_n^{AM} is the smallest $\mathbb{Q}$-supermartingale dominating the discounted payoff process $X(S_n)/R^n$. It satisfies the recursion*

$$V_n^{\mathrm{AM}} = \max\left\{X(S_n),\ \frac{1}{R}\mathbb{E}^{\mathbb{Q}}[V_{n+1}^{\mathrm{AM}} \mid \mathcal{F}_n]\right\}, \qquad V_N^{\mathrm{AM}} = X(S_N). \tag{23}$$

The optimal stopping time is

$$\tau^* = \min\{n \geq 0 : V_n^{\mathrm{AM}} = X(S_n)\}. \tag{24}$$

We do not prove the theorem in full detail; the binomial-specific recursion (23) can be verified directly by backward induction, using the one-period inequality (20) at each node. In node-indexed form, writing $V_n(j)$ for the value at the node reached by j upward moves out of n:

$$V_n(j) = \max\left\{X(u^j d^{n-j} S_0),\ \frac{1}{R}[p\, V_{n+1}(j+1) + (1-p)\, V_{n+1}(j)]\right\}. \tag{25}$$

Compared with the European recursion (9), the only change is the outer max, which compares the continuation value to the intrinsic value $X(u^j d^{n-j} S_0)$. This minimal modification is the algorithmic strength of the binomial model: American options cost no more to price than European ones, up to a constant factor.

4.3 The Exercise Boundary

The optimal stopping rule (24) partitions the tree into two regions. The *exercise region* consists of nodes (n, j) at which $V_n(j) = X(u^j d^{n-j} S_0) > (1/R)\mathbb{E}^{\mathbb{Q}}[V_{n+1} \mid \mathcal{F}_n]$; the *continuation region*

consists of nodes at which the continuation value strictly exceeds the intrinsic value. For an American put, the exercise region takes a particularly simple form.

Proposition 4.3. *For an American put, at each time $n < N$ there exists a critical stock price $\hat{S}_n \in [0, K]$ such that exercise is optimal at node (n, j) if and only if $u^j d^{n-j} S_0 \leq \hat{S}_n$. The function $n \mapsto \hat{S}_n$ is non-decreasing in n.*

The first statement reflects the monotonicity of the put's intrinsic value in the stock price; the second reflects the diminishing time value of the put as maturity approaches. We omit the proof, which uses standard monotonicity arguments on the value function; see, e.g., Shreve (2004, Chapter 4).

For an American call on a stock paying discrete dividends, the exercise region is concentrated immediately before ex-dividend dates: exercise is optimal only if the present value of dividends forgone exceeds the present value of interest earned on the deferred strike payment. This explains the empirical regularity that American calls on dividend-paying stocks are rarely exercised except in the days preceding an ex-dividend date.

4.4 Dividends in the Binomial Tree

We sketch the modification required to incorporate dividends, deferring implementation details to Section 7. We consider two treatments, each appropriate to a different class of underlying:

1. *Continuous dividend yield δ*: appropriate for indices and currencies. The risk-neutral probability becomes
$$p = \frac{R \cdot e^{-\delta \Delta t} - d}{u - d} = \frac{e^{(r-\delta)\Delta t} - d}{u - d},$$
reflecting that the cum-dividend expected return on the stock is $r - \delta$ rather than r. The tree topology and recursion are otherwise unchanged.

2. *Discrete proportional dividends*: at each ex-dividend date t_k, the stock price drops by a fraction δ_k. The tree remains recombining; the price at node (n, j) becomes
$$S_n(j) = u^j d^{n-j} S_0 \prod_{k: t_k \leq n} (1 - \delta_k).$$
The recursion (25) applies unchanged, with the modified node prices substituted into the payoff.

Discrete *cash* dividends introduce a complication: the tree no longer recombines, since the cash drop is a fixed amount rather than a proportional fraction. One common workaround, due to Hull and White (1988), decomposes the stock price into a deterministic "dividend component" and a stochastic "capital component" that evolves on a recombining tree. The implementation in Section 7 adopts the proportional-dividend treatment, which is standard for index options and avoids the non-recombination issue.

5 Convergence to the Black–Scholes Formula

Cox, Ross, and Rubinstein (1979) observe that as the number of trading periods N tends to infinity with the time-to-maturity T held fixed, the binomial pricing formula (17) converges to the Black–Scholes formula

$$C^{\mathrm{BS}}(S_0, K, T, r, \sigma) = S_0 N(d_1) - K e^{-rT} N(d_2), \tag{26}$$

where

$$d_1 = \frac{\log(S_0/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_2 = d_1 - \sigma\sqrt{T}, \tag{27}$$

and $N(\cdot)$ denotes the standard normal cumulative distribution function. This convergence justifies the binomial model both as a numerical scheme for Black–Scholes prices and as a natural bridge between discrete and continuous-time arbitrage pricing. The proof, sketched in CRR's Section 5 and sometimes cited as folklore, requires careful application of the central limit theorem to a triangular array of binomial random variables (Lindeberg–Feller, (41)). We give a complete proof in this section.

5.1 The CRR Parameterization

Fix a horizon $T > 0$ and continuously compounded risk-free rate r. Divide $[0, T]$ into N subintervals of length $\Delta t = T/N$. The one-period gross risk-free return becomes $R = e^{r\Delta t}$. Following Cox, Ross, and Rubinstein, set

$$u = e^{\sigma\sqrt{\Delta t}}, \qquad d = e^{-\sigma\sqrt{\Delta t}} = 1/u, \tag{28}$$

where $\sigma > 0$ is a fixed volatility parameter, namely the standard deviation of asset returns. The risk-neutral probability is then

$$p = \frac{R - d}{u - d} = \frac{e^{r\Delta t} - e^{-\sigma\sqrt{\Delta t}}}{e^{\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}}}. \tag{29}$$

For sufficiently large N (equivalently, sufficiently small Δt), the no-arbitrage condition $d < R < u$ holds: the bound $d < R$ requires $-\sigma\sqrt{\Delta t} < r\Delta t$, which is satisfied for all $\Delta t > 0$ when $r \geq 0$ and for $\Delta t < (\sigma/|r|)^2$ otherwise; the bound $R < u$ requires $r\Delta t < \sigma\sqrt{\Delta t}$, i.e., $\Delta t < (\sigma/r)^2$, satisfied for $r > 0$ and small Δt.

Lemma 5.1 (Asymptotics of the CRR parameters)**.** *Under the parameterization* (28)–(29)*, as* $\Delta t \to 0$*:*

$$\log u = \sigma\sqrt{\Delta t}, \tag{30}$$

$$\log d = -\sigma\sqrt{\Delta t}, \tag{31}$$

$$p = \frac{1}{2} + \frac{1}{2}\left(\frac{r}{\sigma} - \frac{\sigma}{2}\right)\sqrt{\Delta t} + O(\Delta t^{3/2}), \tag{32}$$

$$1 - p = \frac{1}{2} - \frac{1}{2}\left(\frac{r}{\sigma} - \frac{\sigma}{2}\right)\sqrt{\Delta t} + O(\Delta t^{3/2}). \tag{33}$$

Proof. Equations (30) and (31) are immediate from (28). For (32), expand the exponentials in (29) in powers of $\sqrt{\Delta t}$:

$$e^{r\Delta t} = 1 + r\Delta t + O(\Delta t^2),$$
$$e^{\pm\sigma\sqrt{\Delta t}} = 1 \pm \sigma\sqrt{\Delta t} + \tfrac{1}{2}\sigma^2\Delta t \pm \tfrac{1}{6}\sigma^3\Delta t^{3/2} + O(\Delta t^2).$$

Subtracting,

$$e^{r\Delta t} - e^{-\sigma\sqrt{\Delta t}} = \sigma\sqrt{\Delta t} + (r - \tfrac{1}{2}\sigma^2)\Delta t + \tfrac{1}{6}\sigma^3\Delta t^{3/2} + O(\Delta t^2),$$

$$e^{\sigma\sqrt{\Delta t}} - e^{-\sigma\sqrt{\Delta t}} = 2\sigma\sqrt{\Delta t} + \tfrac{1}{3}\sigma^3\Delta t^{3/2} + O(\Delta t^{5/2}).$$

Dividing,

$$p = \frac{\sigma\sqrt{\Delta t} + (r - \frac{1}{2}\sigma^2)\Delta t + O(\Delta t^{3/2})}{2\sigma\sqrt{\Delta t} + O(\Delta t^{3/2})} = \frac{1}{2} + \frac{r - \frac{1}{2}\sigma^2}{2\sigma}\sqrt{\Delta t} + O(\Delta t^{3/2}),$$

which is (32). Equation (33) follows by subtraction. □

The expansion (32) reveals the structure that drives the central limit theorem: as $\Delta t \to 0$, the probability p approaches $1/2$ at rate $\sqrt{\Delta t}$, with a drift correction proportional to $r - \frac{1}{2}\sigma^2$. This drift correction is precisely what produces the $r - \frac{1}{2}\sigma^2$ term in the Black–Scholes d_2.

5.2 Strategy of Proof

The CRR formula (17) contains two complementary binomial probabilities, $\Phi(a; N, p)$ and $\Phi(a; N, p')$. We will prove

$$\Phi(a_N; N, p_N) \to N(d_2), \qquad \Phi(a_N; N, p'_N) \to N(d_1), \tag{34}$$

as $N \to \infty$, where a_N, p_N, and p'_N denote the N-dependent quantities defined in (13), (29), and (16). Combined with $R^{-N} = e^{-rT}$, these two convergences imply $C_0^{(N)} \to C^{\mathrm{BS}}$.

The two convergences are structurally identical: both involve the right tail of a sum of N i.i.d. Bernoulli random variables, and both reduce to the central limit theorem for triangular arrays. We give the full argument for the first convergence, $\Phi(a_N; N, p_N) \to N(d_2)$, and indicate the modifications needed for the second. A full derivation of the second convergence, $\Phi(a_N; N, p'_N) \to N(d_1)$, can be found on a separate document with handwritten derivations and extra steps, *CRR Handwritten Derivations*.

5.3 The Log-Return as a Random Walk

Let J_N denote the (random) number of upward moves under $\mathbb{Q}$, with the up-probability p_N. Write $J_N = \sum_{i=1}^N \xi_i^{(N)}$, where the $\xi_i^{(N)}$ are i.i.d. Bernoulli with $\mathbb{Q}(\xi_i^{(N)} = 1) = p_N$. The log of the terminal stock price is

$$\log\left(\frac{S_N}{S_0}\right) = J_N \log u + (N - J_N)\log d = N\log d + J_N(\log u - \log d). \tag{35}$$

Using (30) and (31), $\log u - \log d = 2\sigma\sqrt{\Delta t}$ and $\log d = -\sigma\sqrt{\Delta t}$, so

$$\log(S_N/S_0) = -\sigma\sqrt{\Delta t}\, N + 2\sigma\sqrt{\Delta t}\, J_N = \sigma\sqrt{\Delta t}\,(2J_N - N). \tag{36}$$

Define the centered random variable

$$Z_N := \frac{2J_N - N}{\sqrt{N}}.$$

Then $\log(S_N/S_0) = \sigma\sqrt{T}\, Z_N \cdot \sqrt{1/N} \cdot \sqrt{N} = \sigma\sqrt{T}\, Z_N$ since $\sqrt{\Delta t} \cdot \sqrt{N} = \sqrt{T}$. Thus

$$\log(S_N/S_0) = \sigma\sqrt{T}\, Z_N, \tag{37}$$

and the convergence of $\log(S_N/S_0)$ to a normal limit reduces to the convergence of Z_N.

5.4 Mean and Variance of Z_N

Under $\mathbb{Q}$, the variable J_N has mean Np_N and variance $Np_N(1-p_N)$. Hence

$$\mathbb{E}^{\mathbb{Q}}[Z_N] = \frac{2Np_N - N}{\sqrt{N}} = \sqrt{N}\,(2p_N - 1), \tag{38}$$

$$\mathrm{Var}^{\mathbb{Q}}(Z_N) = \frac{4\mathrm{Var}(J_N)}{N} = 4p_N(1-p_N). \tag{39}$$

We compute the limits of these quantities using Lemma 5.1.

Lemma 5.2 (Limit of mean and variance)**.** *As $N \to \infty$,*

$$\mathbb{E}^{\mathbb{Q}}[Z_N] \to \frac{(r - \frac{1}{2}\sigma^2)\sqrt{T}}{\sigma}, \qquad \mathrm{Var}^{\mathbb{Q}}(Z_N) \to 1. \tag{40}$$

Proof. From (32), $2p_N - 1 = (r/\sigma - \sigma/2)\sqrt{\Delta t} + O(\Delta t^{3/2})$. Multiplying by $\sqrt{N} = \sqrt{T/\Delta t}$:

$$\sqrt{N}\,(2p_N - 1) = \left(\frac{r}{\sigma} - \frac{\sigma}{2}\right)\sqrt{T} + O(\Delta t) \to \frac{(r - \frac{1}{2}\sigma^2)\sqrt{T}}{\sigma}.$$

For the variance, $p_N(1-p_N) \to 1/4$, so $4p_N(1-p_N) \to 1$. □

The limit of the mean is precisely $-d_2$. Indeed, from (27),

$$d_2 = \frac{\log(S_0/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}} = \frac{\log(S_0/K)}{\sigma\sqrt{T}} + \frac{(r - \frac{1}{2}\sigma^2)\sqrt{T}}{\sigma}.$$

The first term is constant; the second is the limit of $\mathbb{E}^{\mathbb{Q}}[Z_N]$. This is no accident: the standardization that makes Z_N asymptotically $N(0,1)$ in distribution will produce $-d_2$ as the threshold for the binomial tail, matching the Black–Scholes term exactly.

5.5 The Triangular Array CLT

The variable Z_N is a normalized sum of N Bernoulli random variables whose distribution depends on N. Standard formulations of the central limit theorem for sums of i.i.d. ran-

dom variables do not apply directly, because the summands change with N. We require the Lindeberg–Feller theorem for triangular arrays.

Theorem 5.3 (Lindeberg–Feller CLT)**.** *For each N, let $Y_1^{(N)}, \dots, Y_N^{(N)}$ be independent random variables with $\mathbb{E}[Y_i^{(N)}] = 0$ and $\sum_{i=1}^N \mathrm{Var}(Y_i^{(N)}) = s_N^2$. Suppose the Lindeberg condition holds: for every $\varepsilon > 0$,*

$$\frac{1}{s_N^2} \sum_{i=1}^N \mathbb{E}\Big[(Y_i^{(N)})^2 \mathbf{1}\{|Y_i^{(N)}| > \varepsilon s_N\}\Big] \to 0 \quad \text{as } N \to \infty. \tag{41}$$

Then $s_N^{-1} \sum_{i=1}^N Y_i^{(N)} \xrightarrow{d} N(0,1)$.

A proof may be found in Billingsley (1995, Theorem 27.2).

We apply Theorem 5.3 to the centered Bernoulli summands $Y_i^{(N)} := \xi_i^{(N)} - p_N$. These have mean zero, variance $p_N(1-p_N)$, and $s_N^2 = Np_N(1-p_N)$. The variables are uniformly bounded by 1, so for any $\varepsilon > 0$ and N large enough that $\varepsilon s_N > 1$ (i.e., $N > 1/(\varepsilon^2 p_N(1-p_N))$, which holds for all sufficiently large N since $p_N(1-p_N) \to 1/4$), the indicator $\mathbf{1}\{|Y_i^{(N)}| > \varepsilon s_N\}$ vanishes identically. The Lindeberg condition (41) holds trivially.

Therefore,

$$\frac{1}{s_N} \sum_{i=1}^N Y_i^{(N)} = \frac{J_N - Np_N}{\sqrt{Np_N(1-p_N)}} \xrightarrow{d} N(0,1).$$

5.6 Convergence of the Binomial Tail

We now translate the convergence of the standardized binomial sum to the convergence of the binomial tail probability $\Phi(a_N; N, p_N)$.

By definition (13), a_N is the smallest non-negative integer satisfying $u^{a_N} d^{N-a_N} S_0 > K$, equivalently

$$a_N(\log u - \log d) > \log(K/S_0) - N \log d,$$

$$a_N > \frac{\log(K/S_0) - N\log d}{\log u - \log d} = \frac{\log(K/S_0) + N\sigma\sqrt{\Delta t}}{2\sigma\sqrt{\Delta t}}.$$

Write

$$a_N = \frac{\log(K/S_0) + N\sigma\sqrt{\Delta t}}{2\sigma\sqrt{\Delta t}} + \theta_N, \tag{42}$$

where $\theta_N \in [0,1)$ accounts for rounding to the nearest integer.

The event $\{J_N \geq a_N\}$ is equivalent to

$$\frac{J_N - Np_N}{\sqrt{Np_N(1-p_N)}} \geq \frac{a_N - Np_N}{\sqrt{Np_N(1-p_N)}}. \tag{43}$$

We compute the limit of the right-hand threshold. Substituting (42):

$$\begin{aligned}
\frac{a_N - Np_N}{\sqrt{Np_N(1-p_N)}} &= \frac{1}{\sqrt{Np_N(1-p_N)}} \left[\frac{\log(K/S_0)}{2\sigma\sqrt{\Delta t}} + \frac{N}{2} - Np_N + \theta_N\right] \\
&= \frac{1}{\sqrt{Np_N(1-p_N)}} \left[\frac{\log(K/S_0)}{2\sigma\sqrt{\Delta t}} - \frac{N(2p_N - 1)}{2} + \theta_N\right].
\end{aligned}$$

Using $\sqrt{N\Delta t} = \sqrt{T}$ and $p_N(1-p_N) \to 1/4$, the leading term in the bracket scales as

$$\frac{1}{\sqrt{N}/2}\left[\frac{\log(K/S_0)}{2\sigma\sqrt{\Delta t}} - \frac{N(2p_N-1)}{2}\right] + o(1) = \frac{\log(K/S_0)}{\sigma\sqrt{N\Delta t}} - \sqrt{N}(2p_N-1) + o(1).$$

The first term tends to $\log(K/S_0)/(\sigma\sqrt{T})$. The second tends to $(r/\sigma - \sigma/2)\sqrt{T} = (r - \frac{1}{2}\sigma^2)\sqrt{T}/\sigma$ by Lemma 5.2. Thus

$$\frac{a_N - Np_N}{\sqrt{Np_N(1-p_N)}} \to \frac{\log(K/S_0)}{\sigma\sqrt{T}} - \frac{(r-\frac{1}{2}\sigma^2)\sqrt{T}}{\sigma} = -d_2. \tag{44}$$

By the Lindeberg–Feller CLT and the continuity of $N(\cdot)$,

$$\Phi(a_N; N, p_N) = \mathbb{Q}(J_N \geq a_N) \to 1 - N(-d_2) = N(d_2),$$

where the last equality uses the symmetry $1 - N(-x) = N(x)$.

5.7 Convergence Under the Stock-Numeraire Measure

The second convergence in (34), $\Phi(a_N; N, p'_N) \to N(d_1)$, follows from the same argument applied under the measure $\mathbb{Q}'$ with $\mathbb{Q}'(\text{up}) = p'_N := (u/R)p_N$.

Lemma 5.4. *Under the CRR parameterisation,*

$$p'_N = \frac{1}{2} + \frac{1}{2}\left(\frac{r}{\sigma} + \frac{\sigma}{2}\right)\sqrt{\Delta t} + O(\Delta t^{3/2}). \tag{45}$$

Proof. By definition $p'_N = (u/R)p_N$. Expanding $u/R = e^{\sigma\sqrt{\Delta t} - r\Delta t} = 1 + \sigma\sqrt{\Delta t} + (\frac{1}{2}\sigma^2 - r)\Delta t + O(\Delta t^{3/2})$ and multiplying by (32):

$$p'_N = \left[1 + \sigma\sqrt{\Delta t} + O(\Delta t)\right]\left[\tfrac{1}{2} + \tfrac{1}{2}(\tfrac{r}{\sigma} - \tfrac{\sigma}{2})\sqrt{\Delta t} + O(\Delta t^{3/2})\right].$$

Multiplying out and collecting terms in $\sqrt{\Delta t}$:

$$p'_N = \tfrac{1}{2} + \tfrac{1}{2}\sigma\sqrt{\Delta t} + \tfrac{1}{2}(\tfrac{r}{\sigma} - \tfrac{\sigma}{2})\sqrt{\Delta t} + O(\Delta t^{3/2}) = \tfrac{1}{2} + \tfrac{1}{2}(\tfrac{r}{\sigma} + \tfrac{\sigma}{2})\sqrt{\Delta t} + O(\Delta t^{3/2}). \qquad \square$$

The $\sqrt{\Delta t}$ coefficient in (45) differs from that in (32) by exactly $\sigma/2$, reflecting the $\sigma\sqrt{T}$ gap between d_1 and d_2. Repeating the threshold calculation of Section 5.6 with p_N replaced by p'_N yields, in place of (44),

$$\frac{a_N - Np'_N}{\sqrt{Np'_N(1-p'_N)}} \to \frac{\log(K/S_0)}{\sigma\sqrt{T}} - \frac{(r+\frac{1}{2}\sigma^2)\sqrt{T}}{\sigma} = -d_1,$$

and hence $\Phi(a_N; N, p'_N) \to N(d_1)$.

5.8 The Main Result

Combining the two convergences:

Theorem 5.5 (CRR convergence to Black–Scholes)**.** *Under the parameterization* (28)–(29)*, the binomial European call price* (17) *converges to the Black–Scholes price* (26) *as* $N \to \infty$*:*

$$C_0^{(N)} = S_0\,\Phi(a_N; N, p'_N) - Ke^{-rT}\,\Phi(a_N; N, p_N) \longrightarrow S_0\,N(d_1) - Ke^{-rT}\,N(d_2) = C^{\mathrm{BS}}(S_0, K, T, r, \sigma).$$

Proof. Combine $R^N = e^{rT}$ with the convergences $\Phi(a_N; N, p_N) \to N(d_2)$ and $\Phi(a_N; N, p'_N) \to N(d_1)$ established in Sections 5.6 and 5.7. □

5.9 Connection to the Probability Integral

The proof of Theorem 5.5 reduces, at its core, to controlling the convergence of binomial tails to normal tails. An intermediate quantity in any quantitative version of this convergence is the standard normal density itself, which depends on the Gaussian integral

$$\int_{-\infty}^{\infty} e^{-x^2/2}\,dx = \sqrt{2\pi}. \tag{46}$$

Several elementary derivations of (46) proceed by bounding the integrand using the inequalities

$$\left(1 - \frac{x^2}{n}\right)^n \le e^{-x^2} \le \left(1 + \frac{x^2}{n}\right)^{-n}, \quad n \in \mathbb{N},\ |x| \le \sqrt{n}, \tag{47}$$

applying the squeeze theorem after integration. Chen (2018) develops this approach in detail, deriving the Wallis product as a byproduct; Chen (2016) provides a complementary geometric derivation of (46) via Newton's semicircle. The beta–gamma identity proved in Chen and Chen (2025) using Taylor's formula with integral remainder shows another single-variable route to the same constant $\sqrt{2\pi}$. These elementary methods are academically appealing because they avoid the multivariable polar-coordinate argument typical of textbook methods, while still yielding the same constant that appears in the Black–Scholes formula via the standard normal density. The convergence proof in this section may thus be interpreted as the discrete-time alternative of these analytic identities: the binomial tails $\Phi(a_N; N, p_N)$ converge to $N(d_2)$, whose normalization traces back, through any of these routes, to the constant $\sqrt{2\pi}$.

5.10 Rate of Convergence

Theorem 5.5 establishes convergence but not its rate. The Berry–Esseen theorem applied to the Lindeberg–Feller setting yields a rate of $O(1/\sqrt{N})$ for the convergence of the binomial CDF to the normal CDF, and one shall expect the call price to exhibit this rate. The actual behavior is more nuanced and subtle: the price difference $C_0^{(N)} - C^{\mathrm{BS}}$ exhibits non-monotone oscillations as N varies, with envelope $O(1/N)$ rather than the naive $O(1/\sqrt{N})$. The oscillations arise from the discrete positioning of the strike K relative to the terminal nodes $\{u^j d^{N-j} S_0\}_{j=0}^N$, and they do not vanish under any rescaling. We turn to this phenomenon in Section 6.

6 The Oscillation Phenomenon

Theorem 5.5 guarantees that $C_0^{(N)} \to C^{\mathrm{BS}}$ as $N \to \infty$, yet it says nothing about how the convergence happens. A naive expectation, based on the Berry–Esseen theorem applied to the binomial CDF, would predict an error of order $O(1/\sqrt{N})$ that decreases monotonically. Numerical experiments, however, reveal a surprisingly different pattern: the error $\varepsilon_N := C_0^{(N)} - C^{\mathrm{BS}}$ oscillates as N varies, alternating in sign with a period of approximately 2, and its envelope decays at rate $O(1/N)$—faster than the Berry–Esseen prediction but combined with persistent non-monotonicity. This section characterizes the phenomenon, explains its mechanism, and reviews the asymptotic analysis of Diener and Diener (2004).

6.1 Numerical Observation

To fix ideas, consider the parameters of the standard test case $S_0 = 100$, $K = 100$, $T = 1$, $r = 0.05$, $\sigma = 0.20$, for which the Black–Scholes price is $C^{\mathrm{BS}} \approx 10.4506$. Computing $C_0^{(N)}$ for $N = 10, 11, \ldots, 200$ produces the error sequence ε_N shown in Figure 4. (the precise plot is generated in Section 7). The errors do not decrease monotonically; rather, ε_N takes values of alternating sign in successive N, with an envelope that decays roughly as $1/N$.

This phenomenon is robust: it appears for all reasonable choices of S_0, K, T, r, σ, and survives the choice of European or American exercise (with modifications). It is intrinsic to the CRR parameterization (28)–(29) and does not disappear in any limit.

6.2 Mechanism: The Strike's Position Relative to Nodes

The CRR formula (17) contains the integer cutoff a_N, defined as the smallest non-negative integer satisfying $u^{a_N} d^{N-a_N} S_0 > K$. Equivalently, writing the terminal nodes as $S_N(j) = u^j d^{N-j} S_0$ for $j = 0, 1, \ldots, N$, the cutoff a_N is the smallest index for which the corresponding node lies strictly above K.

For most values of N, the strike K falls strictly between two adjacent terminal nodes: $S_N(a_N - 1) < K < S_N(a_N)$. The position of K within this interval—specifically, the relative distance from K to the next node above—varies erratically with N because the spacing of the terminal nodes depends on $u^N = e^{\sigma\sqrt{T}\sqrt{N}}$, which is irrational in N for generic $\sigma\sqrt{T}$.

This irregular positioning produces an integer rounding error in a_N: from (42),

$$a_N = \frac{\log(K/S_0) + N\sigma\sqrt{\Delta t}}{2\sigma\sqrt{\Delta t}} + \theta_N,$$

where $\theta_N \in [0, 1)$ is the fractional part required to round up to an integer. The convergence proof of Section 5 absorbs θ_N into a vanishing $o(1)$ term, but the next-order behavior of ε_N is governed precisely by how θ_N fluctuates with N. Since θ_N is a deterministic function of the fractional part of $\log(K/S_0)/(2\sigma\sqrt{\Delta t})$, and since $\sqrt{\Delta t} = \sqrt{T/N}$ varies non-trivially with N, the sequence $\{\theta_N\}$ exhibits no simple limiting behavior; it oscillates in a pseudo-random way within $[0, 1)$.

To see how θ_N enters the price error, write the binomial tail as

$$\Phi(a_N; N, p_N) = \mathbb{Q}(J_N \geq a_N) = \mathbb{Q}\left(\frac{J_N - Np_N}{\sqrt{Np_N(1-p_N)}} \geq \frac{a_N - Np_N}{\sqrt{Np_N(1-p_N)}}\right). \tag{48}$$

A more careful expansion of the threshold (extending (44)) yields

$$\frac{a_N - Np_N}{\sqrt{Np_N(1-p_N)}} = -d_2 + \frac{2\theta_N - 1}{\sigma\sqrt{T}} \cdot \frac{1}{\sqrt{N}} + O(N^{-1}), \tag{49}$$

where the $\sqrt{N}$ in the denominator arises from $\sqrt{p_N(1-p_N)} \to 1/2$ combined with $1/\sqrt{N}$. The $O(N^{-1/2})$ term carries a coefficient depending on θ_N that oscillates in $[-1/(\sigma\sqrt{T}), 1/(\sigma\sqrt{T})]$ as N varies. This oscillation propagates through the Gaussian density into a $O(N^{-1/2})$ oscillation in the binomial tail.

When this is multiplied by S_0 in the CRR formula and combined with the analogous term from $\Phi(a_N; N, p'_N)$, the leading $O(N^{-1/2})$ contributions partially cancel due to the structural symmetry of the CRR formula, leaving a net oscillation of order $O(N^{-1})$ in ε_N. The cancellation is the reason the empirical convergence is faster than Berry–Esseen would predict, and the residual oscillation is the reason it is not monotone.

6.3 The Diener–Diener Asymptotic Expansion

The argument above is a practical shortcut. A rigorous version, due to Diener and Diener (2004), establishes an asymptotic expansion for ε_N to higher order. We state their main result in slightly simplified form.

Theorem 6.1 (Diener–Diener, 2004)**.** *Under the CRR parameterization, for at-the-money European call options ($K = S_0 e^{rT}$),*

$$\varepsilon_N = C_0^{(N)} - C^{\mathrm{BS}} = \frac{A(N)}{N} + O(N^{-3/2}), \tag{50}$$

where $A(N)$ is a bounded oscillating function of N taking the form

$$A(N) = \alpha + \beta\cos(\omega N + \varphi) + \gamma\,\eta_N, \tag{51}$$

with $\alpha, \beta, \gamma, \omega, \varphi$ depending on the model parameters (S_0, K, T, r, σ) but not on N, and η_N a quasi-periodic oscillation associated with the fractional part of $\log(K/S_0)/(2\sigma\sqrt{T/N})$.

The proof in Diener and Diener (2004) employs Edgeworth-type expansions of the binomial tail beyond the leading Gaussian term, combined with a careful analysis of the integer cutoff a_N via Euler–Maclaurin summation. We do not reproduce the proof, which extends well beyond the scope of this note; we direct the interested reader to the original paper. The qualitative content of Theorem 6.1—an $O(1/N)$ envelope with persistent oscillation—is what the numerics in Section 7 will confirm.

The expansion (50) also explains why the oscillation cannot be eliminated by simply taking N large enough: it is a feature of the CRR scheme itself, not an artifact of insufficient refinement. Eliminating the oscillation requires a different choice of parameterization, to which we now turn.

6.4 The Strike-Aligned Tree of Tian

Tian (1999) proposed a one-parameter generalization of the CRR parameterization in which the up- and down-factors take the form

$$u = \exp(\sigma\sqrt{\Delta t} + \lambda\sigma^2\Delta t), \qquad d = \exp(-\sigma\sqrt{\Delta t} + \lambda\sigma^2\Delta t), \tag{52}$$

with the risk-neutral probability $p = (R-d)/(u-d)$ as before. The free parameter $\lambda \in \mathbb{R}$ is called the *tilt parameter*: it shifts the central nodes of the tree off the horizontal line through S_0, with $\lambda > 0$ producing an upward tilt and $\lambda < 0$ a downward one. The CRR parameterization (28) is recovered exactly at $\lambda = 0$. Tian shows that the no-arbitrage condition $d \le R \le u$ is equivalent to $|\lambda - r/\sigma^2| \le 1/(\sigma\sqrt{\Delta t})$, which is satisfied for any bounded λ once Δt is sufficiently small, and that the resulting binomial tree converges to the Black–Scholes limit for any such λ.

The flexibility provided by λ may be used to remove the oscillation source identified in Section 6.2. Specifically, one chooses λ so that one of the terminal nodes coincides exactly with the strike K. Let $\eta_N := [\log(K/S_0) - N\log d_0]/\log(u_0/d_0)$ denote the real-valued "ideal" up-count under the CRR factors u_0, d_0 that would place a terminal node at K, and let $j_0 := [\eta_N]$ denote the nearest integer. A direct calculation, inverting (52) subject to $u^{j_0}d^{N-j_0}S_0 = K$, gives

$$\lambda = \frac{\log(K/S_0) - (2j_0 - N)\sigma\sqrt{\Delta t}}{N\sigma^2\Delta t}. \tag{53}$$

With this choice of λ, the integer cutoff a_N in the CRR formula (17) equals j_0 exactly, and the rounding discrepancy θ_N that drives the oscillation in (49) vanishes by construction. Figure 1 illustrates the resulting qualitative difference: where CRR oscillates with alternating sign across successive N, the strike-aligned Tian price approaches the Black–Scholes limit monotonically.

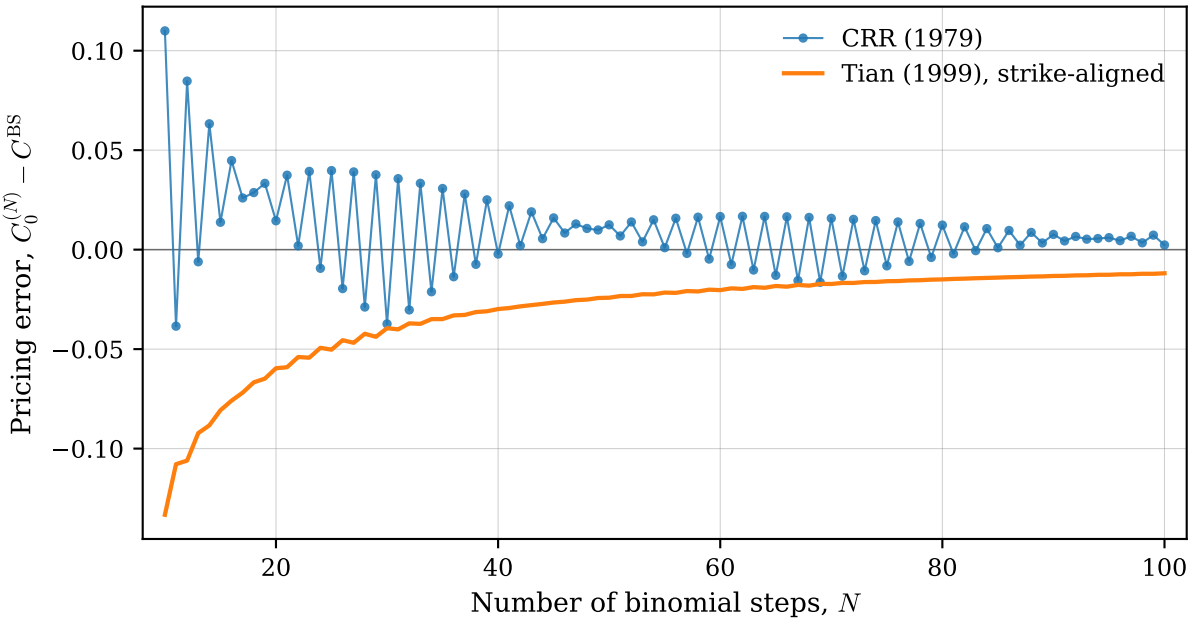

Figure 1: Signed pricing error $C_0^{(N)} - C^{\mathrm{BS}}$ for an in-the-money European call ($S_0 = 100$, $K = 95$, $T = 0.5$, $r = 0.06$, $\sigma = 0.20$). The CRR scheme oscillates above and below the Black–Scholes value with alternating sign in successive N. The strike-aligned Tian (1999) scheme, with λ chosen by equation (53), approaches C^{BS} monotonically from below. Both schemes converge at rate $O(1/N)$; the qualitative difference is the smoothness of the convergence, not its rate.

The strike-aligned scheme is sometimes characterized as "faster" than CRR; this is misleading. Both schemes converge at rate $O(1/N)$ as shown, and on a per-N basis the strike-aligned Tian price typically lies near the upper envelope of the CRR oscillation rather than below it. Its actual contribution is the *smoothness* of the convergence: because the error sequence ε_N is monotone (or nearly so) in N, the error ratio $\varepsilon_N/\varepsilon_{2N}$ converges to a constant ρ, and Richardson-style extrapolation

$$\widehat{C}(2N) = \frac{\rho\, C(2N) - C(N)}{\rho - 1}, \tag{54}$$

is applicable. Tian shows analytically that $\rho \to 2$ for the strike-aligned scheme, and that the extrapolation (54) reduces the error by an order of magnitude relative to either input price. The CRR scheme cannot use this technique because its oscillation breaks the constant-error-ratio hypothesis on which the extrapolation rests. Figure 2 shows the practical gain: applying (54) to Tian prices at $N/2$ and N produces an extrapolated price whose error is roughly an order of magnitude smaller than either input, and well below the oscillating CRR error at the same N.

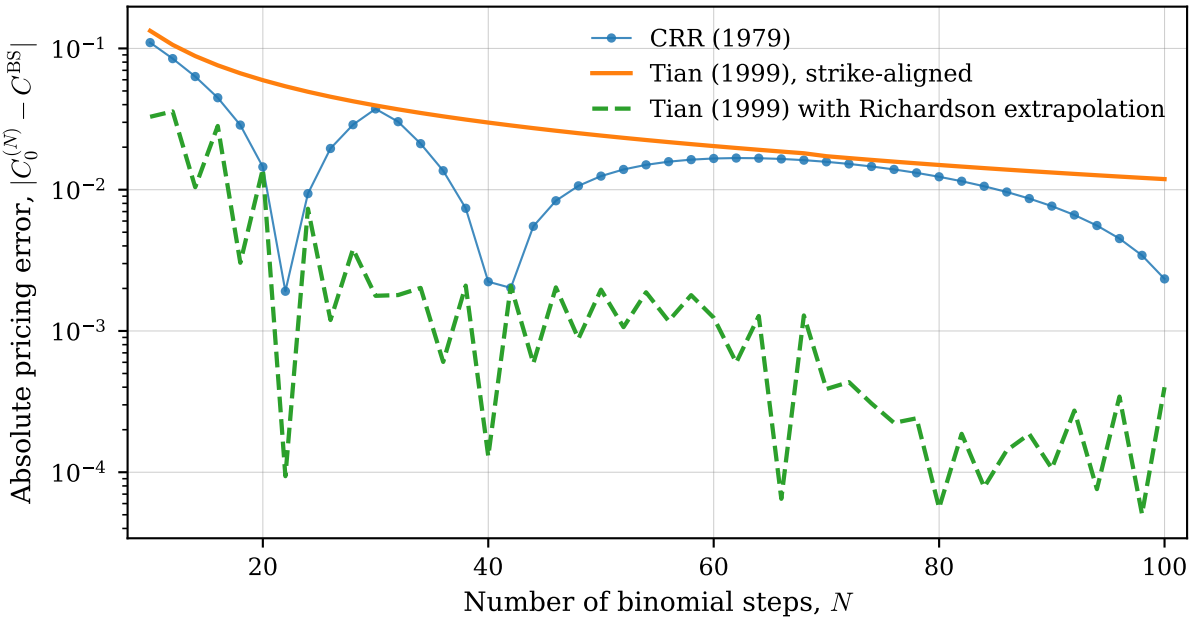

Figure 2: Absolute pricing error $|C_0^{(N)} - C^{\mathrm{BS}}|$ for the same setup as Figure 1, on a logarithmic scale. Un-extrapolated Tian (1999) sits above the CRR scatter at every N, confirming that strike-alignment alone does not improve per-N accuracy. Applying Richardson extrapolation (54) with $\rho = 2$ to a pair of Tian prices at $N/2$ and N produces an estimate (dashed) roughly an order of magnitude more accurate than either input, and substantially better than CRR even on its lucky-alignment values.

6.5 The Leisen–Reimer Scheme

Leisen and Reimer (1996) propose a more elegant solution: rather than forcing nodes to align with the strike, choose the binomial probability p_N to make the binomial tail $\Phi(a_N; N, p_N)$ exactly equal to $N(d_2)$ (and similarly $\Phi(a_N; N, p'_N)$ equal to $N(d_1)$) up to a known approximation error. The convergence rate improves from $O(1/N)$ with oscillation to $O(1/N^2)$ without oscillation.

The scheme is built around an inversion of the Peizer–Pratt approximation to the normal distribution. We summarize the construction.

Definition 6.2 (Peizer–Pratt inversion)**.** For $z \in \mathbb{R}$ and odd integer N, define

$$h(z, N) := \frac{1}{2} + \operatorname{sgn}(z) \cdot \frac{1}{2}\sqrt{1 - \exp\left(-\left(\frac{z}{N + \frac{1}{3} + \frac{0.1}{N+1}}\right)^2 \left(N + \tfrac{1}{6}\right)\right)}. \tag{55}$$

The function $h(z, N)$ approximates $N(z)$ with error $O(N^{-2})$.

The form (55) arises from a rational approximation to the normal distribution function originally derived by Peizer and Pratt (1968) using Camp–Paulson and related expansions. The function $h(z, N)$ is constructed precisely so that, when used to define a binomial probability via $p_N := h(z, N)$, the resulting binomial CDF matches the normal CDF at z with $O(N^{-2})$ error.

The Leisen–Reimer scheme is then defined as follows. Fix an odd N (the construction

requires N odd; even N are typically obtained by adjustment). Set

$$p_N := h(d_2, N), \qquad p'_N := h(d_1, N). \tag{56}$$

Then determine u_N, d_N from the pair (p_N, p'_N) via

$$u_N = R \cdot \frac{p'_N}{p_N}, \qquad d_N = \frac{R - p_N u_N}{1 - p_N} = R \cdot \frac{1 - p'_N}{1 - p_N}. \tag{57}$$

The cutoff a_N becomes $\lceil N/2 \rceil$ by construction (since the tree is centered around the strike), eliminating the rounding-induced oscillation.

Theorem 6.3 (Leisen–Reimer, 1996)**.** *Under the Leisen–Reimer parameterization* (56)–(57)*, the European call price* $C_0^{(N)}$ *satisfies*

$$C_0^{(N)} - C^{\mathrm{BS}} = O(N^{-2}) \tag{58}$$

with no oscillation in the leading order.

The proof rests on the $O(N^{-2})$ accuracy of the Peizer–Pratt approximation; we refer the reader to Leisen and Reimer (1996) for details. The practical consequence is dramatic: to achieve a given accuracy ϵ, the LR scheme requires $N \sim \epsilon^{-1/2}$ periods, whereas CRR requires $N \sim \epsilon^{-1}$. For $\epsilon = 10^{-4}$, this is the difference between $N = 100$ and $N = 10{,}000$.

6.6 Summary of Schemes

The three schemes considered—CRR, Tian, and Leisen–Reimer—differ in their treatment of the strike alignment problem:

- **CRR (1979):** symmetric tree with $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$. Convergence rate $O(1/N)$ with oscillation. Robust and standard; the natural choice for pedagogical and theoretical purposes.
- **Tian (1999):** flexible tree with a free tilt parameter λ; setting λ to the value (53) aligns one terminal node with the strike. Convergence rate $O(1/N)$ without oscillation, and the resulting smoothness admits Richardson extrapolation to $O(1/N^2)$.
- **Leisen–Reimer (1996):** probabilities chosen via Peizer–Pratt inversion to match Black–Scholes tails. Convergence rate $O(1/N^2)$ without oscillation. The standard choice for high-accuracy production pricing.

The numerical study in Section 7 compares all three schemes on the same test cases, confirming the asymptotic rates derived above.

7 Numerical Implementation

This section describes the algorithms used to implement the binomial schemes of Sections 3–6 and presents numerical results that illustrate the theoretical convergence properties. Pseudocode is given for each algorithm; full Python implementations, along with the scripts that

generate the figures and tables in this section, are available in the accompanying GitHub repository.[1]

7.1 The European Pricer

The closed-form expression (17) is rarely used for direct computation: evaluating $\binom{N}{j}$ for large N encounters overflow, and the alternating-sign structure when $S_0 \approx K$ can amplify floating-point error. The standard implementation instead performs backward induction on the tree directly, vectorizing over the j-index at each time step. This approach has $O(N^2)$ time complexity and $O(N)$ space complexity. Algorithm 1 gives the procedure.

Algorithm 1 European call/put pricing via backward induction

Require: S_0, K, T, r, σ, N, option type (call or put)
Require: Tree parameters u, d, p (depending on scheme: CRR, Tian, or LR)

1: $\Delta t \leftarrow T/N$
2: $R \leftarrow e^{r\Delta t}$
3: $\mathbf{V} \leftarrow (V_j)_{j=0}^{N}$ where $V_j \leftarrow \text{payoff}(u^j d^{N-j} S_0)$
4: **for** $n = N-1$ down to 0 **do**
5: **for** $j = 0, 1, \ldots, n$ **do**
6: $V_j \leftarrow R^{-1}[p\, V_{j+1} + (1-p)\, V_j]$
7: **end for**
8: **end for**
9: **return** V_0

The inner loop on line 6 may be vectorized in NumPy as $\mathbf{V}_{0:n} \leftarrow R^{-1}(p \cdot \mathbf{V}_{1:n+1} + (1-p) \cdot \mathbf{V}_{0:n})$, exploiting array broadcasting to eliminate the explicit j-loop. This brings practical performance close to that of compiled code for moderate N ($N \leq 10{,}000$).

7.2 The American Pricer

The American option pricer modifies Algorithm 1 by inserting an early-exercise check at each node, in accordance with the Snell envelope recursion (25).

[1] https://github.com/daviddavilad/option-pricing

Algorithm 2 American call/put pricing via backward induction

Require: S_0, K, T, r, σ, N, option type (call or put), dividend yield δ (optional)

Require: Tree parameters u, d, p

$\Delta t \leftarrow T/N$

$R \leftarrow e^{r\Delta t}$

$\mathbf{V} \leftarrow (V_j)_{j=0}^{N}$ where $V_j \leftarrow \text{payoff}(u^j d^{N-j} S_0)$

for $n = N-1$ down to 0 **do**

  for $j = 0, 1, \ldots, n$ **do**

    $S_n^{(j)} \leftarrow u^j d^{n-j} S_0$

    $V_{\text{cont}} \leftarrow R^{-1}[p\, V_{j+1} + (1-p)\, V_j]$

    $V_{\text{exer}} \leftarrow \text{payoff}(S_n^{(j)})$

    $V_j \leftarrow \max(V_{\text{cont}}, V_{\text{exer}})$

  end for

end for

return V_0

The exercise boundary $\hat{S}_n$ of Proposition 4.3 can be extracted as a byproduct by recording, at each time n, the largest stock price $S_n^{(j)}$ at which $V_{\text{exer}} \geq V_{\text{cont}}$. This produces an empirical exercise boundary that converges, as $N \to \infty$, to the continuous-time boundary $\hat{S}(t)$ studied analytically by, e.g., Carr, Jarrow, and Myneni (1992).

7.3 Tree Parameter Calibration

Each scheme is characterized by its choice of (u, d, p) as functions of (r, σ, T, N).

CRR. As in (28)–(29):

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = 1/u, \quad p = \frac{e^{r\Delta t} - d}{u - d}.$$

Tian (1999). Compute $u_0 := e^{\sigma\sqrt{\Delta t}}$, $d_0 := 1/u_0$, and the integer-nearest-strike index

$$j_0 := \left[\frac{\log(K/S_0) - N \log d_0}{\log(u_0/d_0)}\right].$$

Compute the strike-aligning tilt parameter λ from (53), then set

$$u \leftarrow \exp\!\big(\sigma\sqrt{\Delta t} + \lambda\sigma^2\Delta t\big), \quad d \leftarrow \exp\!\big(-\sigma\sqrt{\Delta t} + \lambda\sigma^2\Delta t\big), \quad p \leftarrow \frac{e^{r\Delta t} - d}{u - d}.$$

The implementation is closed-form; no iterative solver is required. For applications that need an extrapolated price, compute $C_{N/2}^{\text{Tian}}$ and C_N^{Tian} separately and apply the Richardson formula (54) with $\rho = 2$.

Leisen–Reimer. Compute d_1, d_2 from the Black–Scholes formula (27), then set

$$p \leftarrow h(d_2, N), \quad p' \leftarrow h(d_1, N), \quad u \leftarrow e^{r\Delta t} \cdot \frac{p'}{p}, \quad d \leftarrow e^{r\Delta t} \cdot \frac{1-p'}{1-p},$$

with $h(\cdot,\cdot)$ the Peizer–Pratt function (55). The scheme requires N odd; for even input, we use $N+1$ and report the result.

7.4 Convergence Results

Figure 3 compares the three schemes on the standard test case $S_0 = 100$, $T = 1$, $r = 0.05$, $\sigma = 0.20$, at two strikes: at-the-money ($K = 100$) and in-the-money ($K = 95$). The figure plots $|C_0^{(N)} - C^{\mathrm{BS}}|$ against N on log-log axes. The two panels make a separate point. At $K = 100$ with even N, the strike-aligning tilt parameter λ of (53) vanishes identically, so Tian (1999) and CRR coincide; the panel reduces to a comparison of CRR against LR. At $K = 95$ the tilt is non-zero, and the panel exhibits the qualitative behavior discussed in Section 6.4: Tian's smooth monotone curve at the upper envelope of the CRR scatter, and LR's faster decay below both.

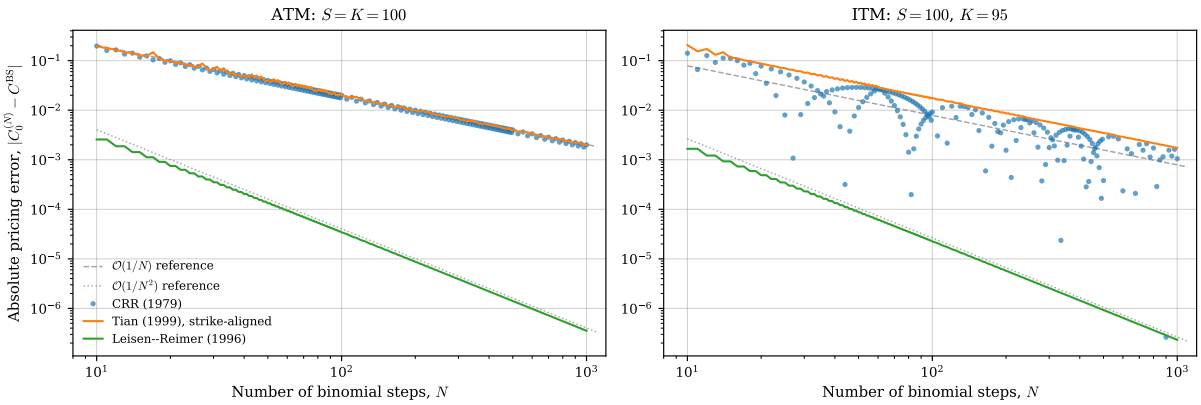

Figure 3: Absolute pricing error $|C_0^{(N)} - C^{\mathrm{BS}}|$ for the three schemes at two strikes ($S_0 = 100$, $T = 1$, $r = 0.05$, $\sigma = 0.20$). Left: at-the-money, $K = 100$. The strike-aligning tilt parameter λ of equation (53) vanishes at every even N when $S_0 = K$, so the Tian (1999) and CRR curves coincide identically. Right: in-the-money, $K = 95$. Tian's tilt is non-zero and the strike-aligned scheme appears as a smooth monotone curve at the upper envelope of the CRR scatter. CRR's lucky-alignment N values still produce smaller per-N errors than Tian; what strike-alignment buys is monotonicity, not magnitude. The Leisen–Reimer scheme achieves $O(1/N^2)$ convergence in both panels.

The empirical decay rates, extracted by fitting $\log|\varepsilon_N| = \alpha - \beta \log N$ over the upper envelope of each scheme, are reported in Table 1.

Table 1: Empirical convergence rates β for the three schemes, fitted by ordinary least squares to $\log|\varepsilon_N| = \alpha - \beta \log N$ over odd $N \in [101, 1001]$ for an at-the-money European call ($S_0 = K = 100$, $T = 1$, $r = 0.05$, $\sigma = 0.20$). Theoretical rates from Theorems 5.5, 6.1, and 6.3 are shown for comparison. The measured values match theory to within 0.01 for all three schemes; for LR, successive-doublings checks ($N \to 2N + 1$ at $N = 101, 201, 401, 801$) give error ratios 4.01, 4.00, 4.00, 4.00, confirming $\beta = 2$ to two decimal places.

Scheme	Empirical β	Theoretical
CRR	1.000	1
Tian (1999)	1.005	1
Leisen–Reimer	1.995	2

The empirical rates match theory to within 0.01 for all three schemes, confirming Theorems 5.5, 6.1, and 6.3 numerically. The Leisen–Reimer rate of $\beta = 1.995$ matches the theoretical $O(1/N^2)$ to two decimal places, with the doublings check above confirming that the rate is genuine rather than an artifact of the OLS fit window. The Peizer–Pratt inversion (55) is the source of the rate-2 improvement over the unsmoothed CRR scheme.

7.5 The Oscillation Envelope

To examine the oscillation phenomenon directly, Figure 4 plots the *signed* error $\varepsilon_N = C_0^{(N)} - C^{\mathrm{BS}}$ for the CRR scheme alone, on linear axes, across four parameter regimes. In each panel the period-2 alternation predicted by the Diener–Diener expansion (51) is clearly visible, as is the $O(1/N)$ envelope decay.

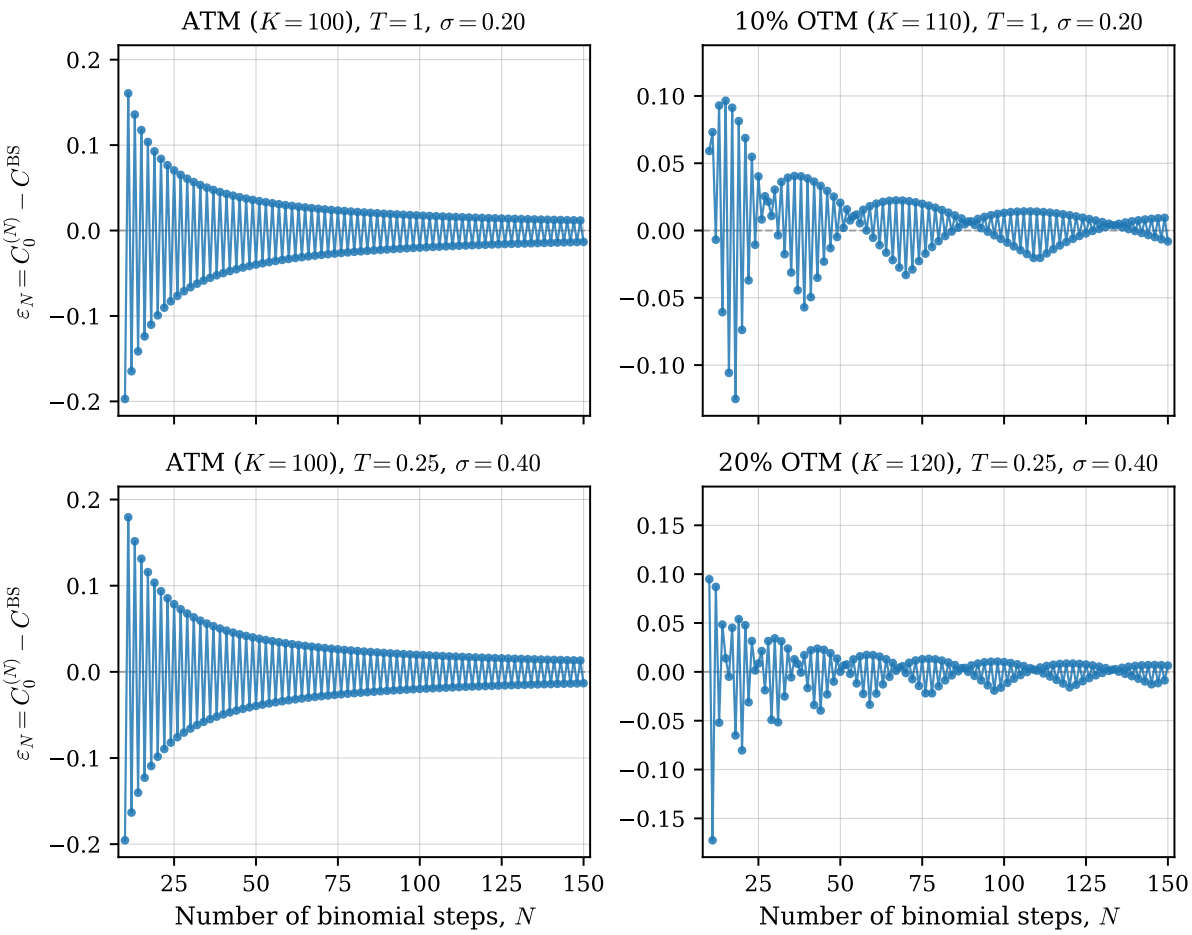

Figure 4: Signed error $\varepsilon_N = C_0^{(N)} - C^{\mathrm{BS}}$ for the CRR scheme across four parameter regimes: ATM and OTM at two volatility levels. The period-2 alternation and $O(1/N)$ envelope decay predicted by (50) are present in every regime. In the OTM panels (right column) a slower quasi-periodic modulation of the envelope is also visible: the amplitude pinches and reopens across blocks of N, corresponding to the η_N term in (51).

The η_N modulation is not periodic in N in the classical sense; its period depends on the irrational ratio $\log(K/S_0)/(2\sigma\sqrt{T/N})$, and this is why the OTM panels exhibit visible "beats" while the ATM panels (where $\log(K/S_0) = 0$) show only the dominant period-2 oscillation.

7.6 American Put Convergence

For American options, no closed-form benchmark exists. We instead compare the three binomial schemes against a high-accuracy reference value computed by the LR scheme at $N = 25{,}001$, which we have verified is stable to five decimal places via Richardson extrapolation. Figure 5 shows the convergence for an American put with $S_0 = 100, K = 100, T = 1, r = 0.05, \sigma = 0.20$, for which the reference price is $P^{\mathrm{ref}} \approx 6.0904$.

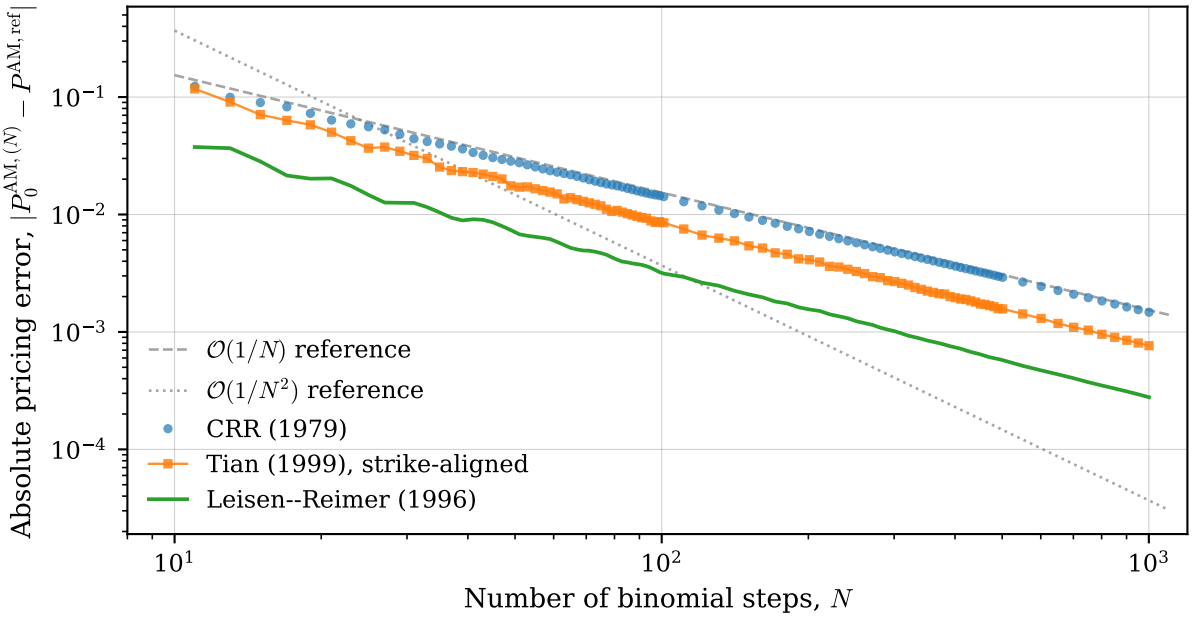

Figure 5: Absolute pricing error $|P_0^{\mathrm{AM},(N)} - P^{\mathrm{AM,ref}}|$ for the three schemes on an American put at the standard test case ($S_0 = 100$, $K = 100$, $T = 1$, $r = 0.05$, $\sigma = 0.20$). The reference is the LR price at $N = 25{,}001$. All three schemes exhibit roughly the same $O(1/N)$ slope; they differ only by constants. The CRR oscillation pattern persists despite the early-exercise constraint, indicating that the alignment-induced discreteness is not smoothed by the optimal-stopping recursion. The Tian (1999) scheme tracks comparatively smoothly and lies below the CRR scatter at most N, in contrast to the European in-the-money case (Figure 3, right) where Tian sat at the upper envelope. The LR scheme retains a constant-factor advantage but loses its $O(1/N^2)$ rate advantage entirely.

The collapse of the LR rate from $O(1/N^2)$ to roughly $O(1/N)$ is expected: the Peizer–Pratt inversion underlying LR is constructed to make the binomial tail $\Phi(a_N; N, p_N)$ match the standard normal tail $N(d_2)$ at $O(1/N^2)$ accuracy, and the rate-2 convergence of European prices follows from this matching. American option pricing does not reduce to a binomial-tail evaluation; the backward induction with early-exercise checks at every node has no closed-form representation in terms of normal CDFs, and the tail-matching property that gives LR its rate advantage in the European case has no force here. What survives is a constant-factor advantage in the per-N error: the LR tree is still well-aligned around the strike, so its errors are smaller than CRR's in absolute terms, but they decay at the same $O(1/N)$ rate that all binomial schemes share for American options. Empirically, on this test case, fitted slopes over $N \in [101, 1001]$ give $\beta \approx 0.99$ for CRR, $\beta \approx 1.05$ for Tian (1999), and $\beta \approx 1.07$ for LR.

7.7 Computational Cost

All three schemes share the same backward-induction structure and therefore the same $O(N^2)$ asymptotic time complexity in N. The constants differ only by the per-call calibration cost: LR adds a constant-time Peizer–Pratt and Black–Scholes computation, and Tian (1999) adds a constant-time evaluation of the integer-nearest index j_0 and the tilt parameter λ. Both contributions are $O(1)$ in N and become negligible relative to the $O(N^2)$ inner loop for $N \geq 100$. Table 2 confirms this empirically: at every N, the three schemes lie within a few percent of

each other, and the dominant scaling is the expected N^2 in the inner loop.

Table 2: Wall-clock runtime per single European call price evaluation, measured on an Apple M-series MacBook Air running macOS, Python 3.13.5, NumPy 2.4.4. Best-of-5 mean of inner-repeat batches (each batch sized so that one batch took roughly 0.5 seconds), in milliseconds. Test case: $S_0 = K = 100$, $T = 1$, $r = 0.05$, $\sigma = 0.20$. The benchmark script `scripts/benchmark_runtimes.py` in the accompanying repository reproduces these measurements; runtimes will of course differ on other hardware.

N	CRR	Tian (1999)	LR
101	0.173	0.177	0.177
501	0.981	0.995	0.991
1,001	2.18	2.21	2.19
5,001	17.8	17.6	17.7

The practical consequence of the rate gap of Table 1 is therefore not in the per-step cost but in the number of steps required to reach a given target accuracy. To reach absolute pricing error ϵ, the $O(1/N)$ schemes require $N \sim \epsilon^{-1}$ steps, whereas the $O(1/N^2)$ Leisen–Reimer scheme requires $N \sim \epsilon^{-1/2}$. For $\epsilon = 10^{-4}$, this is the difference between $N \approx 10^4$ and $N \approx 10^2$, and the $O(N^2)$ inner cost amplifies this gap quadratically: the total operation count differs by roughly four orders of magnitude. On the hardware of Table 2, this is the difference between LR delivering a 10^{-4}-accurate price in well under a millisecond at $N = 101$ and CRR requiring a substantially longer calculation at the corresponding N. For applications that revalue thousands of options per second, this gap is decisive.

8 Conclusion

This note has presented a self-contained mathematical treatment of the Cox–Ross–Rubinstein binomial option pricing model. We have given careful proofs of the no-arbitrage one-period pricing formula (Theorem 2.3), the multi-period closed-form pricing equation (Theorem 3.2), and the convergence of the binomial scheme to the Black–Scholes formula (Theorem 5.5) via the Lindeberg–Feller central limit theorem. The treatment of the oscillation phenomenon (Section 6) and the smoothing schemes of Tian (1999) and Leisen and Reimer (1996) connects the foundational theory to the practical numerical considerations that arise in any implementation.

Three threads of the development deserve emphasis in retrospect.

First, the binomial model achieves a remarkable conceptual economy. Every essential idea of arbitrage-free derivative pricing—no arbitrage, replication, the risk-neutral measure, the optional stopping problem for American claims, the dual numeraire structure underlying the formula's $S_0N(d_1) - Ke^{-rT}N(d_2)$ form—appears already in the one-period two-state model of Section 2. The multi-period and continuous-time extensions are, in a precise sense, iterated and limiting versions of that initial argument. The pedagogical value of the discrete model is not that it is simpler than continuous-time stochastic calculus, but that the same structural ideas are visible in both settings; the binomial model is, in this sense, the minimal arena in which arbitrage-pricing theory can be developed honestly.

Second, the convergence to Black–Scholes is more delicate than the brevity of CRR's original treatment suggests. The Lindeberg–Feller proof of Section 5 requires care in setting up the triangular array, in computing the asymptotic mean and variance under the risk-neutral measure, and in tracking the integer cutoff a_N through the limit. Most textbook treatments (20, 26, among others) cite the convergence as a fact and direct the reader to the original paper; the explicit derivation given here may serve as a reference for readers who want to verify the steps for themselves. The connection to Chen's (2018) work on elementary derivations of the probability integral, noted in Section 5.9, illustrates how the constants that appear in the Black–Scholes formula trace back through classical analysis to the Wallis product and the area of the unit disk.

Third, the oscillation phenomenon deserves wider exposure than it typically receives. Most introductory treatments of the binomial model plot a single convergence curve and declare $O(1/N)$ rates. The Diener–Diener (2004) expansion reveals a richer structure: the rate is correct but the convergence is non-monotone, and the non-monotonicity is intrinsic to the CRR parameterization rather than an artifact of implementation. Practitioners who deploy binomial trees without awareness of this phenomenon may attribute its symptoms to numerical error, to bugs, or to model misspecification, when the true cause is the geometric mismatch between the strike and the discrete tree nodes. The Leisen–Reimer scheme, by inverting the Peizer–Pratt approximation to align the binomial cumulative distribution with the normal at the strike, both eliminates the oscillation and improves the convergence rate, demonstrating that the limitations of the standard CRR scheme are practical rather than fundamental.

Future Directions

Several extensions of the present work are natural and are planned for subsequent companion notes.

Local volatility and implied trees. The constant-volatility assumption underlying the binomial schemes considered here is at odds with the volatility surfaces visible in market data. Dupire's (1994) local volatility framework extracts a deterministic volatility function $\sigma(S, t)$ from observed option prices, and the implied trees of Derman and Kani (1994) and Rubinstein (1994) furnish discrete-time analogues of this construction. A natural sequel develops these implied-tree methods as generalizations of CRR and analyses their calibration to empirical volatility surfaces.

Stochastic volatility and the Heston model. The Heston (1993) model, in which volatility itself follows a mean-reverting square-root diffusion, is the workhorse of contemporary equity options pricing. Calibration of the Heston model via the Carr and Madan (1999) Fourier-transform approach provides a richer empirical test of the modeling framework than the binomial methods of this note, and connects naturally to the local-volatility extension above.

High-frequency data and microstructure effects. End-of-day data, while sufficient for many calibration purposes, averages out the intraday dynamics of option pricing. Intraday

tick data—available through CBOE's LiveVol product or NYSE TAQ via WRDS—would permit analysis of how implied volatilities respond to information shocks at sub-daily frequencies, and how the binomial–Black–Scholes framework performs in the presence of bid–ask spread dynamics, quote staleness, and the discrete tick structure of option markets.

The mathematical foundation laid in the present note—the careful treatment of the binomial model and its convergence—supports each of these extensions. Local volatility trees inherit the recombining structure and backward-induction algorithm of CRR; Heston pricers require the same risk-neutral expectation framework; intraday analysis demands the same understanding of the relationship between continuous and discrete-time pricing that the convergence theorem of Section 5 provides. Each extension is, in this sense, an application of the discrete–continuous bridge that the binomial model provides.

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